

Paper_ED_GW_01 — Black-Hole Ringdown Spectroscopy as Time-Varying Saddle-Hessian-Signature Reconfiguration

Series: Wave-3 Generative Papers (Gravitational-Wave Arc — direct extension of Paper_ED_GW_00) **Status:** Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED)** within DCGT A→regime. Direct composition of Dyn_01 (S2 mixed-signature saddle framework) + GW_00 (NoetherFlux propagation chain) + Route A4 substrate-parameter inheritance; no new primitives or paper-specific postulates. **Date:** 2026-05-17 **Authors:** Paper_087 (13 primitives); Paper_073 (DCGT); Paper_089/090/093 (V1/V5/T18 kernels); Paper_012 (substrate-c); Paper_027 (Newton's G); Paper_039 (horizon decoupling); Paper_047 (Hawking via V5 KMS); Paper_ED_SC_4_9 (S1/S2/S3); Paper_ED_GW_00 (M3 + row 12 partial); Paper_ED_Dyn_01 (M3 — S2 regime); Paper_ED_Dyn_02 (M2 — horizon motion); Paper_ED_Dyn_03 (M2 — radiation law); Paper_ED_SC_4_1 (BH cosmic decoupling); Paper_095 (verdict grammar — §3.2.1 sub-labels). Substrate-research support memos cross-referenced in §2.

Label-class note: Audit-row sub-labels *substrate-parameter-INHERITED* and *IC-INHERITED* per Paper_095 §3.2.1.

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim a closed-proof derivation of standard general-relativistic Quasi-Normal Mode (QNM) spectroscopy from substrate primitives. The continuum-side QNM machinery (Regge-Wheeler equation; Teukolsky equation; spin-weighted spheroidal harmonics) is **INHERITED** from standard linearized GR via DCGT continuum bridge. ED supplies the substrate-graph reading of *what* propagates and *what* decays during ringdown — not a substrate-graph rederivation of standard linearized-GR machinery.
2. It does **not** sit within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). ED is in the substrate-ontology research program ('t Hooft cellular-automaton; Wolfram Ruliad; causal-set program, Sorkin et al.). Methodology per Paper_095 (form-FORCED / value-INHERITED).
3. It does **not** introduce new substrate primitives or paper-specific postulates. The ringdown content is composition of Paper_ED_GW_00 propagation framework + Paper_ED_Dyn_01 saddle-dynamics framework + Route A4 substrate-parameter-INHERITED scale + standard linearized-GR QNM machinery via DCGT translation.
4. It does **not** claim numerical-precision derivation of specific QNM frequencies $\omega_{lmn}(M, J)$ for given BH parameters. Specific QNM-frequency numerical values are INHERITED from the standard Schwarzschild / Kerr QNM spectrum at continuum-side; ED supplies the substrate-graph mechanism that *produces* the spectrum, not a substrate-graph numerical recomputation. Substrate-graph derivation of $G + c +$ BH mass content is via Paper_027 + Paper_012 + standard astrophysics inheritance, respectively.
5. It does **not** posit a fundamental metric near the horizon. Per Paper_ED_GW_00 §3.2, substrate has no fundamental metric to perturb; ringdown is the decay of *substrate-graph saddle-Hessian-signature reconfiguration* to the equilibrium signature pattern, not the decay of metric perturbations on a Kerr background.
6. It does **not** address ringdown phenomenology beyond the substrate-graph identification + first-order mode-spectrum content. Higher-order overtones, mode-mixing across (l, m) mul-

tipoles, and nonlinear ringdown phenomena (spectroscopy at high SNR, “no-hair theorem” tests beyond fundamental modes) inherit standard astrophysics + standard QNM analysis at continuum side; ED’s substrate-graph reading supplies foundational mechanism, not a parallel rederivation of these refinements.

7. **Inheritance from Dyn_02 M2 status:** GW_01 inherits Paper_ED_Dyn_02’s BH-horizon-as-mixed-signature-saddle identification + horizon-motion framework. Dyn_02’s OPEN-HM-Q1 + OPEN-HM-Q2 (chain-class criterion + stress-energy form per regime) remain OPEN at Dyn_02’s level; GW_01 does not load-bear on these closures because the ringdown-specific substrate-graph mechanism (Hessian eigenstructure $\rightarrow$ discrete modes + V5 finite-memory damping) is identified independently of which chain-class content underlies non-saturation regimes. Same architectural-division logic as Dyn_01 + Dyn_04.

Abstract

Black-hole ringdown is identified at substrate-graph level as **the decay of time-varying saddle-Hessian-signature reconfiguration in the near-horizon mixed-signature substrate region to the equilibrium horizon signature pattern**. The mechanism inherits from Dyn_01 (saddle dynamics) applied to the BH horizon as a mixed-signature (S2-class) saddle locus per Dyn_02; outward propagation via GW_00’s V1-retarded NoetherFlux chain. **Form-IDENTIFIED** via Dyn_01 + GW_00 + Dyn_02 composition. Mode quantization arises from substrate-action Hessian eigenstructure: $\omega_n = \omega_R^{(n)} + i\omega_I^{(n)}$ with $\omega_R^{(n)} \sim \sqrt{|\text{Re } \lambda_n|}$ (kernel-induced curvature) and $\omega_I^{(n)} \sim 1/\tau_{V_5}^{\text{eff}}$ (V5 finite-memory dissipation). **Value-INHERITED:** $\omega \sim c^3/(GM)$ via standard astrophysics + Paper_027; $\tau_{V_5} + \Theta_{\text{ED}}$ at substrate-parameter-INHERITED level via Route A4. **IC-INHERITED** from NR-merger phenomenology + Dyn_02. Substrate-c bound preserved throughout. **Verdict per Paper_095: M3. F1 (sharpest):** ringdown-mode structure inconsistent with mixed-signature saddle-Hessian eigenstructure (e.g., $\omega \approx 1/M$ at leading order; or damping rate inconsistent with substrate-side τ_{V_5}) refutes the composition. Enables Paper_ED_GW_02 (stochastic background); supplies the ringdown component of Paper_ED_GW_00 row 13 closure (M3 ringdown / M2 full-waveform via Dyn_05 mixed-tier).

§1 Statement of Result

BH ringdown is the substrate-graph **decay of time-varying mixed-signature saddle-Hessian-signature reconfiguration to the equilibrium horizon signature pattern**. The substrate-side identification composes:

1. **BH horizon as mixed-signature (S2-class) saddle locus** (Dyn_01 §3.6 + Dyn_02): the substrate-action Hessian near the BH horizon has compression-dominant + expansion-dominant orthogonal axes (mixed signature per SC-4.9 S2 class).
2. **Time-varying saddle-Hessian-signature reconfiguration** as the post-merger dynamical content (Dyn_01 §3.7): merger perturbs the equilibrium signature pattern; decay back to equilibrium is the ringdown.
3. **V1 retarded propagation at substrate-c** carries the decaying signature-reconfiguration outward as gravitational waves (GW_00 §3.3 + §3.9 row 12 partial closure).

4. **Mode quantization from substrate-graph Hessian eigenstructure** near the horizon: complex eigenfrequencies $\omega_n = \omega_R^{(n)} + i\omega_I^{(n)}$ of the time-varying saddle eigenproblem.
5. **Characteristic scale** $\omega \sim c/r_{\text{horizon}} = c^3/(2GM)$ inherits standard BH-mass astrophysics; substrate-parameter content inherits from Route A4 at substrate-parameter-INHERITED level.

Three temporal regimes (the standard BH-merger phenomenology, substrate-side reading):

Regime	Substrate-graph content	Continuum-side phenomenology
Pre-merger (inspiral)	Two mixed-signature saddle loci with slowly time-varying coupled signature patterns	Standard inspiral GW emission per Dyn_03 quadrupole formula (GW_00 row 12)
Plunge (signature merging)	Two saddle loci merge to single highly-perturbed mixed-signature saddle	Strongly nonlinear transition; standard numerical-relativity domain
Ringdown	Decay of perturbed single saddle's signature pattern to equilibrium	Damped oscillations at QNM frequencies ω_{lmn} ; standard QNM spectroscopy

FORM: ringdown = time-varying saddle eigenstructure decay — **IDENTIFIED** via GW_00 + Dyn_01 + Dyn_02 composition. Zero pure-D rows.

VALUE: $\omega \sim c^3/(GM)$ inherits standard astrophysics; substrate-parameter $\tau_{V_5} + \Theta_{\text{ED}}$ at substrate-parameter-INHERITED level (Paper_027-analog) via Route A4.

IC: Hessian-signature configuration at the post-merger BH locus inherited from horizon-motion (Dyn_02) + numerical-relativity merger phenomenology (continuum-side inheritance for specific initial perturbation amplitudes).

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime per §4 audit table.

§2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02 (participation), P03 (translation invariance — Noether source), P09 (polarity), P10 (multiple rule-types — V1, V5 kernel structure), P11 (commitment / time arrow), P12 (ED-threshold + stability landscape — Hessian-signature relevance), P13 (translation arrow).

Upstream papers: - Paper_012 (P-RB-1 substrate-c bound) - Paper_027 (Newton's G — substrate-parameter inheritance for GM content) - Paper_073 (DCGT hydrodynamic-window coarse-graining) - Paper_089 (V1 retarded kernel + T18 forward-causal) - Paper_090 (V5 cross-chain finite-memory — sets damping rate $1/\tau_{V_5}$) - Paper_093 (T18 kernel-arrow direction) - Paper_039 (BH horizon as substrate decoupling surface) - Paper_047 (Hawking spectrum at $T_H = \kappa/(2\pi)$ via V5 KMS — substrate-graph BH content) - Paper_ED_SC_4_9 (saddle-geometry classification S1/S2/S3; mixed-signature S2 class corresponds to BH horizon) - Paper_ED_GW_00

(GW propagation; row 12 partial form-closure via NoetherFlux substrate-graph chain — supplies propagation framework for ringdown radiation outward) - Paper_ED_Dyn_01 (saddle dynamics — universal generator framework; ringdown = time-varying mixed-signature saddle case) - Paper_ED_Dyn_02 (horizon-motion law — supplies near-horizon dynamical framework; mixed-signature regime at saturation regime row) - Paper_ED_Dyn_03 (radiation law — supplies quadrupole / Larmor formulas for inspiral phase) - Paper_ED_SC_4_1 (BH horizon — cosmic decoupling — cross-arc identification at substrate-parameter level) - Update_ED_SC_4x_Arc_M3 (six-projection arc-wide M3 upgrade post-Route-A; BH horizon projection) - Standard linearized-GR QNM machinery (Regge-Wheeler / Teukolsky equations; spin-weighted spheroidal harmonics; complex eigenfrequency formalism) — textbook inheritance via DCGT continuum bridge

Substrate-research support: - Memo_ED_RouteA_MultiRouteConvergence_Audit (multi-route convergence option (ii) at substrate-parameter-INHERITED level) - Memo_ED_Q1Q2_JointClosure_Const + Memo_ED_ChainClass_C3_Construct + Audit (Q1A + Q2A + C3 chain for saturation-regime context; mixed-signature non-saturation context inherits Dyn_02's M2 status for chain-class identification)

No new substrate primitives. No paper-specific postulates.

§3 Derivation

§3.1 Saddle-Hessian structure near horizon

Per Paper_ED_Dyn_02 + Paper_ED_GW_00 §3.1 corpus statement (“the horizons are the dynamics”): a BH horizon is a substrate-graph mixed-signature saddle locus — the substrate-action Hessian at the horizon has compression-dominant axes (along the radial direction inward to the BH interior — substrate-action concavity along inward-radial) and expansion-dominant axes (along outward-radial direction — substrate-action convexity). This is the standard substrate-graph reading of the BH horizon as the boundary between causally-connected (exterior) and causally-disconnected (interior) substrate content.

The mixed-signature saddle is **stable** at equilibrium: the Hessian eigenstructure at equilibrium gives the static Schwarzschild / Kerr horizon configuration. Small perturbations to the signature pattern (e.g., from a post-merger transient) decay back to equilibrium under V1 + V5 + DCGT dynamics — this decay is the ringdown.

§3.2 Time-varying signature reconfiguration during merger

During a binary BH merger, two mixed-signature saddle loci merge into a single highly-perturbed mixed-signature saddle. The post-merger configuration is *not* at the equilibrium Hessian-signature pattern; it carries time-varying signature reconfiguration content that decays toward equilibrium over the ringdown timescale.

Per Paper_ED_Dyn_01 §3.7 (time-varying saddle-Hessian-signature reconfiguration — radiation + GW propagation): the time-varying signature content carries non-trivial Noether flux T_{sub}^{0i} outward via V1 retarded propagation (Paper_089) at substrate-c (Paper_012 P-RB-1). DCGT (Paper_073) translates this substrate-graph flux to continuum gravitational-wave radiation. The framework is identical to GW_00's row 12 partial form-closure mechanism (Paper_ED_GW_00 §3.9 + Noether-Flux substrate-graph chain), applied to the specific case of near-horizon time-varying signature

configuration.

§3.3 Mode quantization from substrate-graph Hessian eigenstructure

The substrate-action Hessian near the perturbed-equilibrium BH saddle has eigenstructure

$$\mathcal{H}\vec{u}_n = \lambda_n \vec{u}_n$$

where $\vec{u}_n$ are substrate-graph eigenmodes of the near-horizon Hessian. Each eigenmode supports a discrete frequency of oscillation about the equilibrium pattern:

$$\omega_n = \omega_R^{(n)} + i\omega_I^{(n)}$$

with $\omega_R^{(n)} \sim \sqrt{|\text{Re } \lambda_n|}$ (oscillation frequency from kernel-induced curvature scale; see Paper_ED_Dyn_01 §3.2 + Memo_ED_RouteA_A2_Construct §2.2 IR-limit dispersion analysis) and $\omega_I^{(n)} \sim 1/\tau_{V_5}^{\text{eff}}$ (damping rate from V5 finite-memory dissipation; $\tau_{V_5}^{\text{eff}}$ is the effective V5 memory in the near-horizon regime, scale-matched to the local Hessian content).

The discrete-mode structure is the substrate-graph analog of standard QNM spectrum quantization; the substrate-graph eigenproblem inherits its functional content from standard linearized-GR Regge-Wheeler / Teukolsky machinery via DCGT continuum bridge.

§3.4 Relation to GW_00 propagation law

Outward propagation of the decaying signature reconfiguration is GW_00's NoetherFlux chain verbatim: substrate-side Noether flux $T_{\text{sub}}^{0i} \rightarrow \text{DCGT} \rightarrow \text{continuum } P_{\text{GW}} = (G/5c^5)\langle\ddot{Q}_{ij}\ddot{Q}^{ij}\rangle$ (GW_00 §3.9 row 12 partial form-closure). For ringdown, $\ddot{Q}_{ij}$ oscillates at QNM frequencies ω_n with damping $\omega_I^{(n)}$; outward radiation inherits the same oscillation-with-damping structure. Speed = substrate-c (GW170817 constraint $|c_{\text{GW}} - c|/c < 10^{-15}$); two transverse polarizations (GW_00 §3.7); no scalar / vector modes.

§3.5 Relation to Dyn_02 horizon motion

Paper_ED_Dyn_02 supplies the substrate-graph dynamical framework for horizon motion; ringdown's near-horizon time-varying signature reconfiguration is a specific instance — the BH horizon itself evolves during ringdown toward Kerr equilibrium via mixed-signature saddle dynamics per Dyn_02 + Dyn_01. Post-merger initial-condition supply: standard numerical-relativity merger phenomenology supplies specific perturbation amplitudes; substrate-graph framework supplies the structural identification of *what* is perturbed (the mixed-signature saddle-Hessian-signature pattern).

§3.6 Value inheritance from Route A

Per Paper_ED_Dyn_01 §3.4 + Memo_ED_RouteA_MultiRouteConvergence_Audit option (ii): the substrate-parameter-INHERITED scale is $\ell_{\text{saddle}} \sim \ell_P(\Theta_{\text{ED}}^{\text{Planck-units}})^{-1/2}$ at saturation; reduces to $\ell_{V_5} = c\tau_{V_5}$ in kernel-dominated IR regime. For BH ringdown, the relevant scale is the near-horizon Hessian-eigenvalue scale composed with $r_{\text{horizon}} = 2GM/c^2$. QNM frequency scaling $\omega \sim c^3/(2GM)$ inherits standard astrophysics; substrate-parameter $\tau_{V_5}/\tau_P \approx 10^{61} +$

$\Theta_{\text{ED}}^{\text{Planck-units}} \approx 10^{-122}$ at Paper_027-analog inheritance tier. Tightening via RA-OPEN-4a/4b/4c-explicit is substrate-research-frontier, not blocking.

§3.7 Comparison to standard GR QNMs

Element	Standard GR QNM	ED substrate-graph reading	Status
What oscillates	Metric perturbation $h_{\mu\nu}$ on Kerr background	Saddle-Hessian-signature pattern at substrate-graph level (no fundamental metric per GW_00 §3.2)	Replaced (substrate-ontology change)
Mode quantization origin	Eigenvalue problem of Regge-Wheeler / Teukolsky equation on Kerr background	Substrate-action Hessian eigenstructure near horizon (§3.3)	Identified (structurally equivalent; substrate-graph mechanism replaces metric-perturbation mechanism)
Frequency scaling $\omega \sim 1/M$	Geometric, from $r_{\text{horizon}} \sim GM/c^2$	Geometric, from ℓ_{saddle} at near-horizon Hessian-eigenvalue scale; inherits r_{horizon} via standard astrophysics	Inherited (numerical) + Identified (mechanism)
Damping rate	From energy-loss to outgoing GW + ingoing-horizon flux	From V5 finite-memory dissipation τ_{V_5} + V1 propagation outward + ingoing-horizon Hessian-signature equilibration	Identified + substrate-graph mechanism replaces standard horizon-flux argument
Spectrum content (no-hair)	Determined by (M, J) alone for stationary BH	Determined by equilibrium Hessian-signature pattern, which depends on (M, J) via standard BH astrophysics inheritance	Inherited (no-hair structural feature)

Element	Standard GR QNM	ED substrate-graph reading	Status
Higher overtones / mode mixing	Standard QNM analysis	Higher Hessian eigenmodes (substrate-graph); inherits standard QNM machinery via DCGT continuum bridge	Inherited
Substrate UV cutoff content	None (continuous spacetime)	$\omega \rightarrow c/\ell_P$ regime at edge of DCGT validity (Q-A4-3-analog regime restriction); not load-bearing at standard astrophysical frequencies	Identified + edge-of-validity for substrate-scale frequencies

§4 Audit Table

#	Step	Label	Notes
1	P02, P03, P09, P10, P11, P12, P13	P	Paper_087. P03 + P13 supply Noether translation invariance; P12 supplies Hessian-signature relevance.
2	Upstream inheritance block	I	Paper_012; Paper_027; Paper_039; Paper_047; Paper_073 (DCGT A→regime); Paper_089; Paper_090; Paper_093; Paper_ED_SC_4_9; Paper_ED_SC_4_1; Paper_ED_GW_00; Paper_ED_Dyn_01; Paper_ED_Dyn_02; Paper_ED_Dyn_03; Up-date_ED_SC_4x_Arc_M3.

#	Step	Label	Notes
3	Standard linearized-GR QNM machinery (Regge-Wheeler / Teukolsky / spheroidal harmonics)	I	Textbook inheritance via DCGT continuum bridge; no substrate-graph rederivation.
4	BH horizon = mixed-signature (S2-class) saddle locus	D-via-I	§3.1. Dyn_01 + Dyn_02 + SC-4.9 S2 + GW_00 §3.1 composition.
5	Equilibrium-stability of mixed-signature saddle at static Schwarzschild / Kerr horizon	D-via-I	§3.1. SC-4.9 Hessian + standard BH thermodynamics.
6	Post-merger time-varying saddle-Hessian-signature reconfiguration	D-via-I	§3.2. Row 4 + Dyn_01 §3.7.
7	Mode quantization from substrate-graph Hessian eigenstructure: $\omega_n = \omega_R^{(n)} + i\omega_I^{(n)}$	D-via-I	§3.3. Row 4 + Dyn_01 §3.2 + Memo_RouteA_A2 §2.2 IR-dispersion.
8	$\omega_R^{(n)} \sim \sqrt{ \text{Re } \lambda_n }$ from kernel-induced curvature	D-via-I	§3.3. Inherits Dyn_01 §3.2 + §3.4.
9	$\omega_I^{(n)} \sim 1/\tau_{V_5}^{\text{eff}}$ from V5 finite-memory dissipation	D-via-I	§3.3. Paper_090 + Dyn_01 + near-horizon scale-matching.
10	V1 retarded propagation outward at substrate-c carries decaying signature reconfiguration	D-via-I	§3.4. Paper_089 + Paper_012 + GW_00 §3.9 NoetherFlux chain.
11	Continuum GW radiation $P_{\text{GW}} \propto \ddot{Q}_{ij} ^2$ at ringdown timescale	D-via-I	§3.4. GW_00 §3.9 + Dyn_03 quadrupole.
12	Two transverse polarizations ($h_+, h_\times$); no scalar / vector / longitudinal modes	I	GW_00 §3.7 + Paper_109 + Paper_116 inheritance.

#	Step	Label	Notes
13	Speed of GW propagation = substrate-c (GW170817 constraint)	I	GW_00 §3.8 + Paper_012 inheritance.
14	Characteristic frequency scaling $\omega \sim c^3/(2GM) = c/r_{\text{horizon}}$	I	§3.6. Standard astrophysics + Paper_027.
15	$\tau_{V_5} + \Theta_{\text{ED}}$ at Route A4 substrate-parameter-INHERITED level	I	Sub-type per Paper_095 §3.2.1: <i>substrate-parameter-INHERITED</i> . §3.6. Memo_RouteA_MultiRouteConverge option (ii); Paper_027-analog tier.
16	Post-merger Hessian-signature configuration (IC) inherited from NR-merger phenomenology + Dyn_02	I	Sub-type per Paper_095 §3.2.1: <i>IC-INHERITED</i> . §3.5. NR supplies specific amplitudes; substrate-graph framework supplies what is perturbed.
17	Substrate-c bound preserved throughout	D-via-I	§3.4. Paper_012 + Paper_089 + Paper_093.
18	Verdict M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime	A→position	Per Paper_095 §3.3 line 77. Five-anchor verdict-sync: status / abstract / §1 / this row / §6 all M3.

Audit summary: zero pure-D rows; **nine substantive D-via-I rows** (4–11, 17); five I rows (2, 3, 12, 13, 14); one substrate-parameter-INHERITED row (15); one IC-INHERITED row (16); one P row (1); one A→position verdict row (18). **No load-bearing OPEN items.** All upstream OPEN content (Dyn_02/Dyn_03 chain-class identifications via Routes C1/C4; SC-4.9 substrate-FORCED exhaustiveness; Route A tightening RA-OPENs) is inherited at upstream papers' status; GW_01 does not depend on those OPENs for its M3 verdict. No new substrate primitives; no paper-specific postulates.

§5 Falsification Criteria

- **F1 (sharpest):** Observation of ringdown-mode structure in high-SNR LIGO/Virgo/KAGRA / future Einstein Telescope / Cosmic Explorer data inconsistent with mixed-signature saddle-Hessian eigenstructure. Specifically: mode-frequency scaling significantly deviating from $\omega \sim 1/M$ at leading order (beyond standard astrophysics + numerical-relativity uncertainty) refutes the substrate-graph $\ell_{\text{saddle}} \sim r_{\text{horizon}}$ identification at §3.6. Currently consistent: observed ringdown frequencies across BH-BH merger catalog (GW150914 and successors) match standard QNM-frequency-scaling predictions at observational precision.
- **F2:** Detection of ringdown damping rates inconsistent with $\omega_I^{(n)} \sim 1/\tau_{V_5}^{\text{eff}}$ scaling at substrate-parameter-INHERITED level — e.g., observation of significantly longer-lived ringdown than substrate-graph V5 finite-memory prediction allows. Refutes §3.3 + audit row 9. Currently consistent: observed ringdown timescales match standard QNM damping-rate predictions at observational precision.
- **F3:** Detection of non-transverse polarization modes (scalar / vector / longitudinal) in ringdown data beyond observational uncertainty. Refutes Paper_ED_GW_00 §3.7 inheritance (audit row 12). Currently consistent: LIGO/Virgo polarization-content analyses constrain non-tensor modes.
- **F4 (structural form):** Substrate-graph proof that mixed-signature saddle Hessian eigenstructure does not admit discrete-mode quantization with the predicted scaling — e.g., a substrate-graph no-go theorem for near-horizon Hessian discrete-eigenmode structure. Substrate-research-frontier counterargument level; **empirical refutation not currently available** (the empirical analog — high-SNR observation of a continuous, non-discrete ringdown spectrum — is essentially F1 with different framing and currently consistent). Refutes §3.3 + audit row 7.

§6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), not within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

BH ringdown in ED is the decay of time-varying mixed-signature saddle-Hessian-signature reconfiguration in the near-horizon substrate region to the equilibrium horizon signature pattern, propagating outward via Paper_ED_GW_00's V1-retarded NoetherFlux chain. Mode quantization arises from substrate-graph Hessian eigenstructure near the horizon — complex eigenfrequencies $\omega_n = \omega_R^{(n)} + i\omega_I^{(n)}$ with $\omega_R^{(n)}$ from kernel-induced curvature and $\omega_I^{(n)}$ from V5 finite-memory dissipation. Frequency scaling $\omega \sim c^3/(GM)$ inherits standard astrophysics; substrate-parameter content $\tau_{V_5} + \Theta_{\text{ED}}$ at substrate-parameter-INHERITED level via Route A4. The substrate-ontology distinguishing content vs standard GR: mixed-signature saddle reconfiguration *replaces* metric perturbation on Kerr background; no graviton, no metric perturbation, no Hilbert-mode (GW_00 §3.7 inheritance).

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime hydrodynamic-window per Paper_095. Nine D-via-I rows; five I rows;

one substrate-parameter-INHERITED row (Route A4); one IC-INHERITED row (NR-merger phenomenology + Dyn_02); zero pure-D rows; no paper-specific postulates; no new substrate primitives; **no load-bearing OPEN items** (upstream Dyn_02 OPEN-HM-Q1/Q2 inherited at upstream status per Preamble item 7; not load-bearing for GW_01). Standard QNM phenomenology (empirically supported by GW150914 and successors) is structurally inherited; substrate-ontology framing is the genuine ED contribution.

Cross-arc consequences. Upstream, GW_01 inherits unchanged from GW_00 (M3), Dyn_01 (M3), Dyn_02 (M2 — chain-class OPENs not aggravated, see Preamble item 7), Dyn_03 (M2), Paper_ED_SC_4_1 (strengthens — BH horizon now substrate-graph-derived as mixed-signature saddle at substrate-parameter-INHERITED level), and the ED-SC 4.x arc-wide M3 picture. Downstream, GW_01 *enables* Paper_ED_GW_02 (Stochastic GW Background) draft via composition with GW_00 + standard cosmology (mixed-tier per Paper_095 §3.2.2: ringdown component M3 via GW_01; inspiral component M2 via Dyn_05 inheritance); *strengthens* Paper_ED_Dyn_04 by supplying the post-collapse ringdown component; *strengthens* Paper_ED_Dyn_05 by supplying the post-merger ringdown of the full-waveform framework; and *closes* Paper_ED_GW_00 row 13 (full inspiral-merger-ringdown waveform; standing OPEN) at **mixed-tier per Paper_095 §3.2.2: M3 for the ringdown component (GW_01); M2 for the full inspiral-merger-ringdown waveform composition (inherits Dyn_05's M2 base verdict for the inspiral component)**. The minimum-tier-rule of composition makes the full-waveform closure M2 even though GW_01 itself sits at M3.

End Paper_ED_GW_01.